

SCOUT REPORT - R MARKDOWN WORKFLOW

README / usage overview

Purpose: Summary of how the R Markdown file loads the prepared scouting environment, selects players, and generates the automated visual report.

R Markdown | FBref-derived database | Understat shot data | Automated visual report

1. Core idea

The Rmd is the report generator. It does not recreate the main player database. It loads `Funcions Scout Report.Rdata`, which is expected to contain the prepared player dataset (`DATA`) and the reusable plotting functions. After loading the required packages, the user changes three parameters and the Rmd calls the visualisation functions in sequence.

Critical dependency: `Funcions Scout Report.Rdata` must be present and loaded before the report functions can run.

2. Main parameters

```
JUGADOR <- "Marcus Rashford"
JUGADOR2 <- "Anthony Gordon"
POSICIO <- "FW,MF"
```

- `JUGADOR`: focal player used in the individual plots and shotmap.
- `JUGADOR2`: comparison player used in the two-player radars.
- `POSICIO`: positional group used to define the scatterplot comparison sample.

The Rmd recommends checking `DATA` first to verify exact player spelling and position labels.

3. Automatic report sequence

Setup - Load `Funcions Scout Report.Rdata`, inspect `DATA`, and attach the required R packages.

Radars - Generate four radar outputs: individual/comparative and raw/percentile.

Barplots - Generate a category-based percentile profile and a top-8 percentile plot.

Scatterplot - Automatically select the focal player's two strongest eligible metrics and compare similar-position players.

Shotmap - Download team-season shot data from Understat and create the focal player's shot map.

4. Required files / data

- `Funcions Scout Report.Rdata` - saved environment with `DATA` and functions.
- Prepared player database used by the functions.
- Internet access for the Understat section.

5. Visualisations generated

5.1 Radar charts

- ``create_raw_radar_1(DATA, JUGADOR)`` - individual raw per-90 radar.
- ``create_raw_radar_2(DATA, JUGADOR, JUGADOR2)`` - raw two-player comparison.
- ``create_perc_radar_1(DATA, JUGADOR)`` - individual percentile radar.
- ``create_perc_radar_2(DATA, JUGADOR, JUGADOR2)`` - percentile two-player comparison.

The radars organise variables into shot creation, passing, dribbling and defence. The Rmd states that these radars do not apply a positional filter. Percentile radars rank players on a 0-100 scale and therefore remove the original metric units.

5.2 Barplots

- ``create_barplot_radar(DATA, JUGADOR)`` - percentile bars organised by performance category.
- ``create_top8_barplot(DATA, JUGADOR, n_metriques = 8)`` - the player's eight highest percentile metrics.

These plots are intended to make strengths and weaker areas faster to identify than in the radar format.

5.3 Automatic scatterplot

```
create_scatterplot_best_metrics(DATA, JUGADOR, POSICIO, usar_minuts = TRUE)
```

The function calculates percentiles, finds the focal player's two strongest eligible metrics, uses them as the x and y axes, and filters the comparison sample using ``POSICIO``.

- ``usar_minuts = TRUE``: point size represents minutes played.
- ``usar_minuts = FALSE``: point size uses a non-standardised category reference variable.
- The focal player and leading players on each axis are highlighted/labelled.

5.4 Shotmap

```
understat_team_meta("Barcelona")  
link <- "https://understat.com/team/Barcelona/2024"  
shotsunder <- understat_team_season_shots(link)  
create_shotmap_resultat(shotsunder, JUGADOR)
```

The Understat team and URL must be changed to match the player's team. The shotmap filters the downloaded shots to ``JUGADOR``, uses colour for shot result and point size for xG, and reports summary statistics such as shots, goals, conversion and xG.

6. How to generate another report

- 1. Open the Rmd in the same project directory as `Funcions Scout Report.Rdata`.
- 2. Load/confirm the saved environment contains `DATA` and the required functions.
- 3. Change `JUGADOR`, `JUGADOR2` and `POSICIO`.
- 4. Change the Understat team and season URL for the shotmap section.
- 5. Knit/render the Rmd to HTML or PDF.

7. Functions called directly by the Rmd

Radar: create_raw_radar_1, create_raw_radar_2, create_perc_radar_1, create_perc_radar_2

Barplots: create_barplot_radar, create_top8_barplot

Scatterplot: create_scatterplot_best_metrics

Shotmap: create_shotmap_resultat

8. Interpretation / scope

The Rmd describes the report as a preliminary scouting and consultation tool rather than a complete player evaluation. Per-90 standardisation, possession adjustments and percentile conversion should be considered when interpreting players with unusual roles, limited minutes or substitute-heavy usage.

9. Historical reproducibility note

Archived data workflow: The reporting stage can be preserved using the frozen/prepared database even though the original FBref scraping workflow is no longer operational.

Before publishing the repository, verify one inconsistency in the current materials: the Rmd introduction states a minimum of 500 minutes, while the functions script loads a file named `5BIGLEAGUES\_400m.rds`. The README should ultimately state the threshold actually used by the frozen dataset.

10. Minimal repository structure

```
scout-report/  
  README.pdf  
  Funcions Scout Report RMD.Rmd  
  Funcions Scout Report.Rdata  
  SCOUT REPORT FUNCIONS.R  
  5BIGLEAGUES_400m.rds  
  archived_preprocessing/
```

Based on the current Scout Report R Markdown and function script